

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Examination – Summer 2024

Course: F.Y.B. Tech.

Branch: Common to All Branches

Sem: I

Subject: Engineering Chemistry

Subject code: BTBS102

Marks: 60

Date: 02/ 07/ 2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Draw a neat diagram wherever necessary
3. Figures to right indicates full marks

	(Level /CO)	Marks
Q.1 Solve Any TWO of the following.		
A) Explain the determination of hardness of water by EDTA method.	2	06
B) Discuss disadvantages of hard water in Domestic and Industrial use.	1	06
C) Explain in detail Ion exchange process of softening of water with its advantages and disadvantages.	2	06
Q.2 Solve Any TWO of the following.		
A) Explain Phase diagram of two component Ag-Pb alloy system.	2	06
B) Discuss the term Phase & Component involved in Phase Rule equation with examples.	1	06
C) Explain the Phase diagram of one component Water system.	2	06
Q.3 Solve Any TWO of the following.		
A) Describe in brief Direct (Dry) chemical corrosion.	1	06
B) Discuss Hydrogen evolution mechanism involved in electrochemical corrosion.	2	06
C) Explain Cathodic protection method to minimize the rate of corrosion.	2	06
Q.4 Solve Any TWO of the following.		
A) Write a note on: Refining of Petroleum.	1	06
B) What are Fuels? How are they classified? State the characteristic of good fuel.	2	06
C) Discuss Physical properties of lubricants.	2	06
Q.5 Solve Any TWO of the following.		
A) Describe Wheatstone's bridge method for conductance measurement of a solution.	2	06
B) Describe Ostwald's theory of Acid-Base indicator.	1	06
C) Describe Conductometric titration with any two examples.	2	06

***** End of Question Paper *****